

SYNOPSIS

STUDENT NAME:

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PROJECT TITLE:

SkyBook Flight Booking & Revenue Analytics

1. Introduction

Sky Book – Flight Booking & Revenue Analytics is an end-to-end cloud-based analytics project designed to transform flight booking data into meaningful business insights. The project combines Amazon Web Services (AWS), SQL, and Microsoft Power BI to demonstrate a complete data journey from cloud storage to interactive visualization.

The workflow begins by storing the flight dataset in Amazon S3. AWS IAM is used for access and role management. AWS Glue discovers the structure of the data and registers the metadata in the Glue Data Catalog. Amazon Athena is then used to perform SQL-based analysis on the cataloged flight data. The data is connected to Power BI through Athena ODBC, prepared using Power Query, and calculated using DAX measures. The final output is the Sky Book interactive dashboard.

The project focuses on booking volume, revenue, airline performance, routes, fares, passenger information, payment methods, booking status, monthly trends, and city-level revenue.

2. Problem Statement

Flight booking datasets can contain many records covering different airlines, routes, fares, cities, payment methods, dates, and booking statuses. Looking at raw records alone makes it difficult to quickly identify important patterns and business information. A structured cloud-to-dashboard workflow is therefore useful for storing, preparing, analysing, and presenting the data in an understandable form.

Sky Book addresses this need by combining cloud data services, SQL analysis, and interactive business intelligence. Instead of depending only on raw tables, the project provides a visual way to understand booking and revenue performance.

3. Objectives

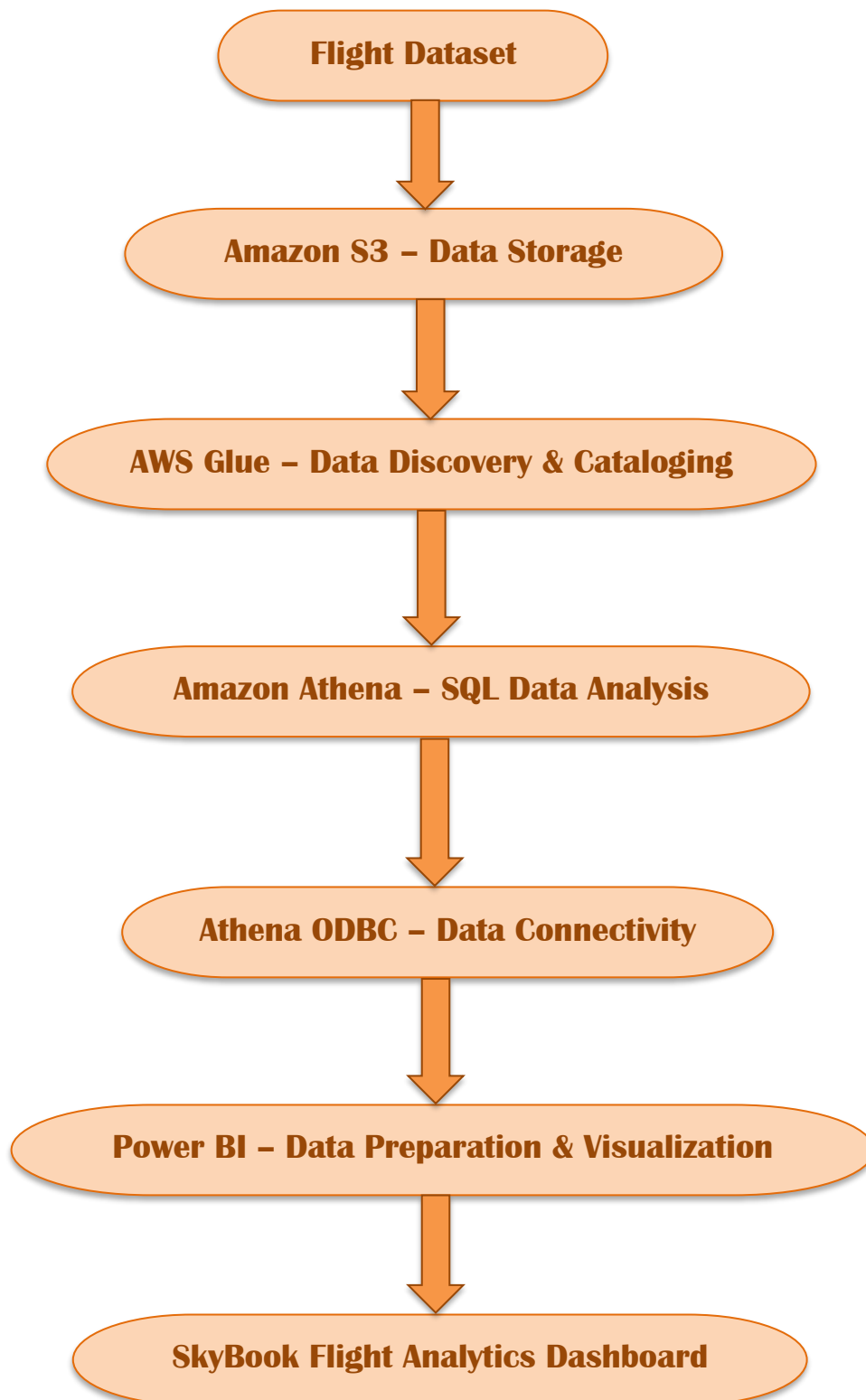
- Implement an end-to-end cloud-based workflow for flight booking data.
 - Store the flight dataset in Amazon S3.
 - Manage service access using AWS IAM.
 - Use AWS Glue and the Glue Crawler for schema discovery and data cataloging.
 - Use Amazon Athena and SQL for flight booking and revenue analysis.
 - Connect the cataloged data to Power BI through Athena ODBC.
 - Prepare and validate the data using Power Query.
 - Create dynamic KPIs and calculations using DAX.
 - Develop an interactive Power BI dashboard with overview and detailed analytics pages.
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4. Technologies and Services Used

Technology / Service	Purpose
Amazon S3	Cloud storage for the flight dataset
AWS IAM	Access and role management
AWS Glue	Data discovery and schema-related processing
Glue Data Catalog	Metadata and table registration
Amazon Athena	Serverless SQL analysis
SQL	Flight booking and revenue analysis
Athena ODBC	Connectivity between Athena and Power BI
Power Query	Data preparation and transformation
DAX	KPI and business calculations
Power BI	Interactive dashboard development

5. System Workflow

The complete implementation follows the sequence:



The workflow separates the main responsibilities of the solution. S3 provides cloud storage, IAM manages access, Glue discovers and catalogs the data, Athena provides SQL analysis, and Power BI presents the results through interactive visuals.

6. AWS Data Preparation

Amazon S3

The flight booking file is uploaded to the project S3 location. The actual source bucket used in the implementation is new-etl-source-bucket, with the flight data located under the configured project folder.

AWS IAM

The IAM role new-etl-role is used to provide the required permissions for the Glue process to access the S3 data and work with the Glue Data Catalog.

AWS Glue and Crawler

A Glue database named customer-db is used for the flight dataset. The crawler new-etl-crawler is configured to inspect the S3 data, identify its available fields and data types, and register the resulting table information in the Glue Data Catalog.

7. Data Analysis Using Amazon Athena

After the flight data is cataloged, Amazon Athena is used as the SQL analysis layer. The analysis covers core booking metrics, airline and route performance, booking status, passenger age, payment methods, departure cities, city-level revenue, fare ranking, and confirmed, cancelled, and completed bookings.

Examples of Analytical Questions

- Which airlines have higher booking volumes?
- Which routes have more bookings?
- How are bookings distributed by status?
- Which payment methods are used?
- Which departure cities contribute more revenue?

A set of 20 SQL queries was designed around the fields and analysis represented in the Sky Book dashboard. The queries support the business questions addressed by the report.

8. Power BI Data Preparation

After the AWS-side analysis, the cataloged flight data is connected to Power BI through the configured Athena ODBC connection. Power BI Desktop is used to access the required database and table.

Power Query is then used to check and prepare important fields such as flight date, fare, age, airline, route, city, booking status, and payment method. The date field is prepared for monthly trend analysis.

9. DAX and KPI Development

DAX measures are created to calculate the key indicators used in the dashboard. These measures provide dynamic values that respond to report filters.

- Total Bookings – total number of booking records.
 - Total Revenue – total fare value.
 - Average Fare – average fare.
 - Average Age – average passenger age.
 - Confirmed Bookings – confirmed booking records.
 - Cancelled Bookings – cancelled booking records.
 - Completed Bookings – completed booking records.
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10. Dashboard Overview

Page 1 – Flight Booking Overview

The first page answers the basic question: “What is happening with the flight bookings?” It brings the most important information into one view so users can understand the overall situation quickly.

- Total Booking – overall booking volume.
- Total Passenger – passenger count represented in the report.
- Total Revenue – overall revenue generated.
- Total Airline – number of airlines represented.
- Total Routes – number of routes represented.
- Bookings by Airline – compares booking volume across airlines.

This page moves from the overall picture toward airline, payment, time-based, and detailed booking views.

Page 2 – Revenue & Booking Analytics

The second page answers the question: “Where is the booking and revenue performance coming from?”

- Average Fare – typical fare value in the dataset.
- Average Passenger Age – overall passenger-age view.
- Confirmed, Cancelled and Completed Bookings – booking outcomes.
- Revenue Analysis – revenue performance.

Together, the two pages create a simple analytical story: Page 1 gives the overall picture, and Page 2 provides deeper insight into revenue and booking performance.

11. Benefits of the Project

- Provides a complete cloud-to-dashboard analytics workflow.
 - Stores the flight dataset in a cloud-based storage layer.
 - Uses automated schema discovery through AWS Glue.
 - Supports SQL-based analysis through Amazon Athena.
 - Reduces the need to manually inspect large numbers of raw records.
 - Provides interactive filtering and visual analysis through Power BI.
 - Helps compare airlines, routes, cities, payment methods, fares, and booking status.
 - Supports faster understanding of booking and revenue performance.
 - Provides a workflow that can be extended for larger or additional flight datasets.
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12. Conclusion

Sky Book demonstrates how cloud services, SQL, data preparation, and business intelligence can be combined into one practical analytics solution. The project starts with raw flight booking data in Amazon S3 and moves through IAM, AWS Glue, Glue Data Catalog, Amazon Athena, SQL, Athena ODBC, Power Query, and DAX before reaching the final Power BI dashboard.

The final dashboard transforms the underlying flight records into a clear visual story covering booking volume, airline performance, revenue, fares, routes, payment methods, booking status, passenger information, monthly trends, and city-level revenue.

“From raw flight data to meaningful business insight.”